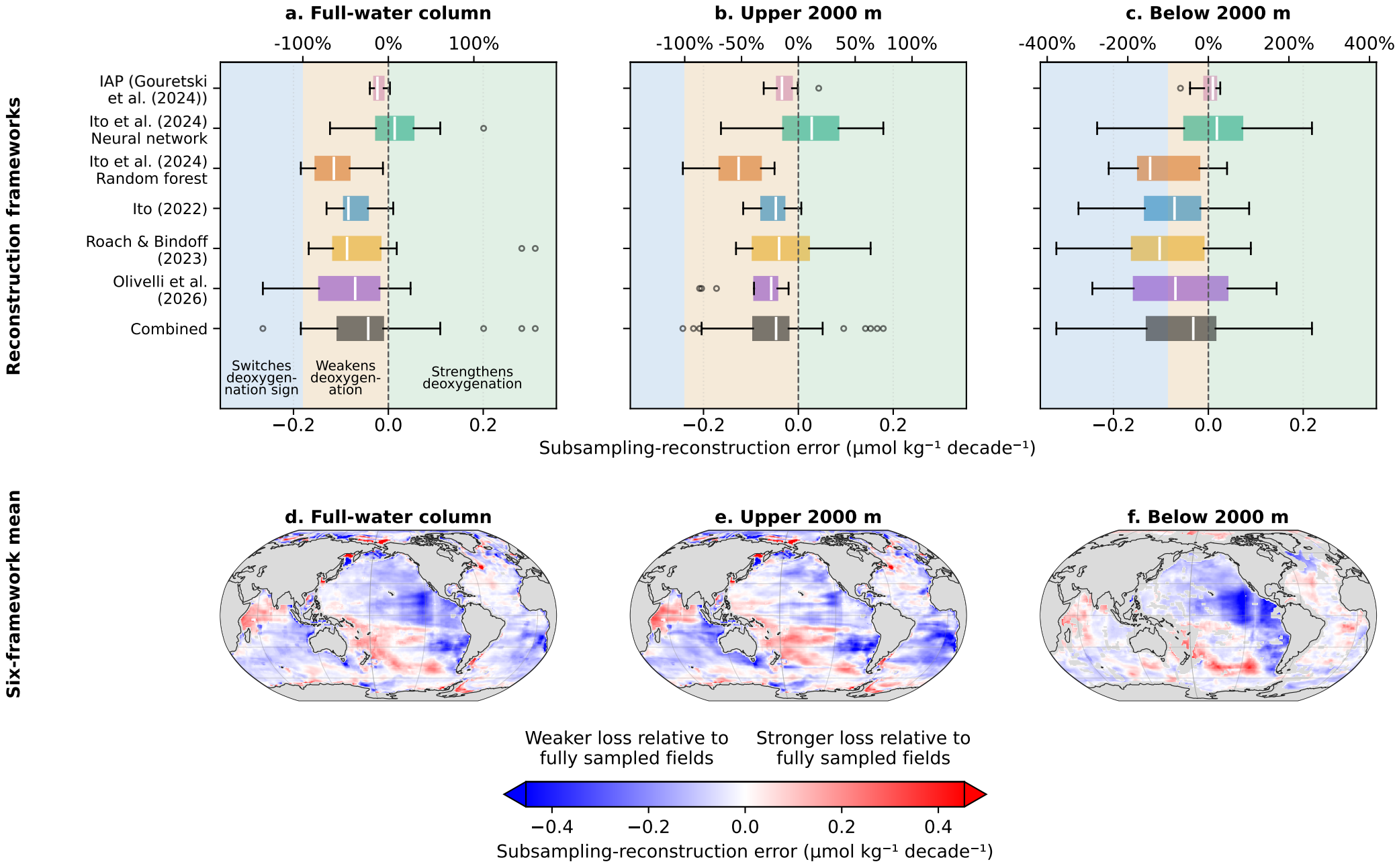

Reconstruction frameworks
a. Full-water column
b. Upper 2000 m
c. Below 2000 m
-100%
0%
100%
-100%
-50%
0%
50%
100%
-400%
-200%
0%
200%
400%
IAP (Gouretski et al. (2024))
Ito et al. (2024) Neural network
Ito et al. (2024) Random forest
Ito (2022)
Roach & Bindoff (2023)
Olivelli et al. (2026)
Combined
Switches deoxygen-nation sign
Weakens deoxygen-ation
Strengthens deoxygenation
−0.2
0.0
0.2
Subsampling-reconstruction error (µmol kg⁻¹ decade⁻¹)
Six-framework mean
d. Full-water column
e. Upper 2000 m
f. Below 2000 m
Weaker loss relative to fully sampled fields
Stronger loss relative to fully sampled fields
−0.4
−0.2
0.0
0.2
0.4
Subsampling-reconstruction error (µmol kg⁻¹ decade⁻¹)